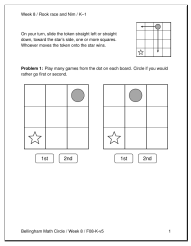
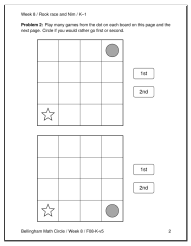
Week 8 / Rook race and Nim / Problem menu

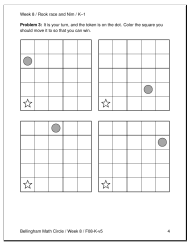
for adults: pick, then print



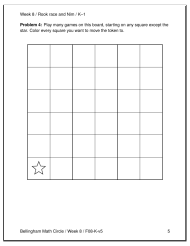
Two 3-by-3 boards: first or second?
K–1 P1
first route



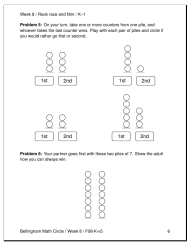
Four 4-by-4 boards: first or second?
K–1 P2
first route



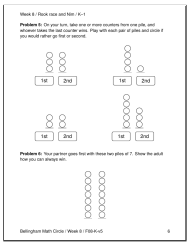
The winning move on four 5-by-5 boards
K–1 P3
return route



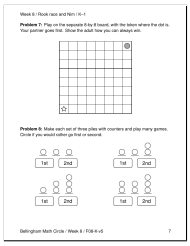
Every square to move to on a 6-by-6 board
K–1 P4
return route



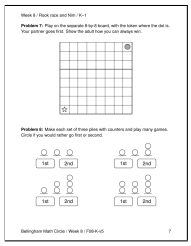
Two piles: first or second?
K–1 P5
first route



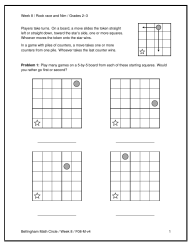
Two piles of 7: show you can always win
K–1 P6
first route



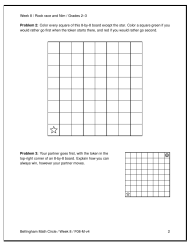
The 8-by-8 board from the far corner
K–1 P7
return route



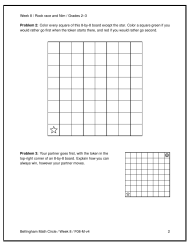
Three piles: first or second?
K–1 P8
return route



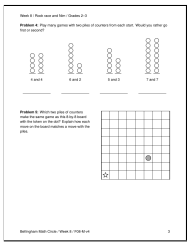
Four starts on a 5-by-5 board
Grades 2–3 P1
first route



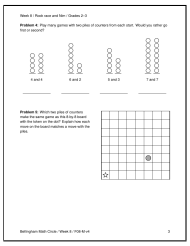
Colour the 8-by-8 board
Grades 2–3 P2
first route



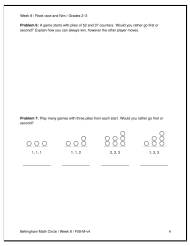
Explain the second player's win from the corner
Grades 2–3 P3
first route



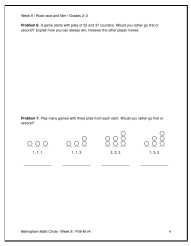
Two piles: first or second?
Grades 2–3 P4
first route



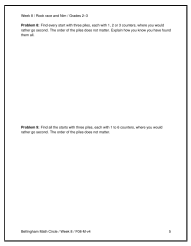
The board as two piles
Grades 2–3 P5
first route



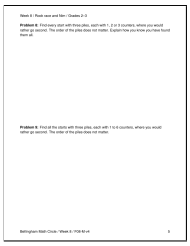
Piles of 52 and 37
Grades 2–3 P6
return route



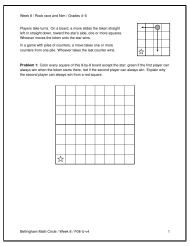
Small three-pile games
Grades 2–3 P7
return route



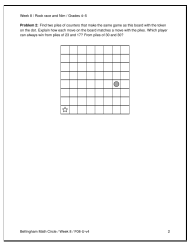
Every second-player start with piles of 1 to 3
Grades 2–3 P8
return route



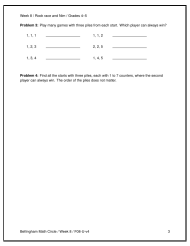
Every second-player start with piles of 1 to 6
Grades 2–3 P9
return route



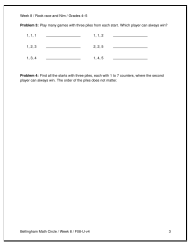
Colour the 8-by-8 board and explain the red squares
Grades 4–5 P1
first route



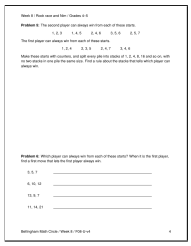
The board as two piles; 23 and 17, 30 and 30
Grades 4–5 P2
return route



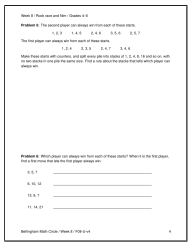
Six three-pile starts
Grades 4–5 P3
first route



Every second-player start with piles of 1 to 7
Grades 4–5 P4
first route



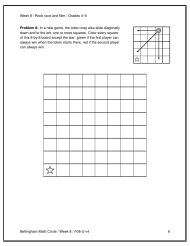
A rule from stacks of 1, 2, 4, 8
Grades 4–5 P5
first route



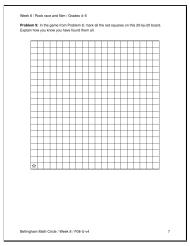
Who wins, and a winning first move
Grades 4–5 P6
first route



Why the stack rule always works
Grades 4–5 P7
return route



The queen game on the 8-by-8 board
Grades 4–5 P8
return route



The queen game on a 20-by-20 board
Grades 4–5 P9
return route